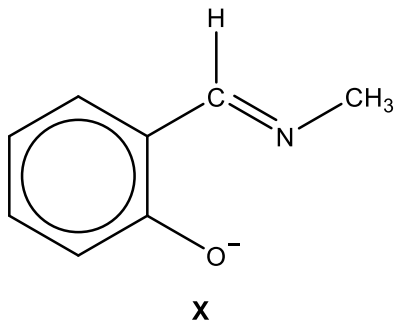
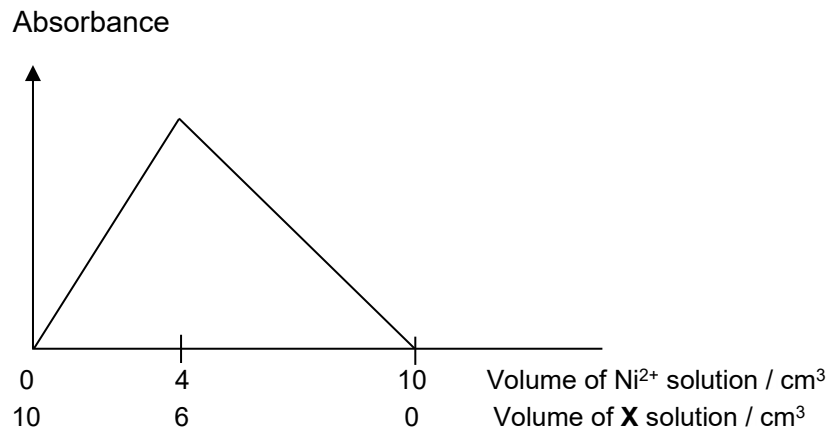


- 29** The complex of nickel with ligand **X** is thermochromic, being coloured red at room temperature but changing to yellow-green when heated to 170 °C.



The following graph sketch was obtained when the absorbances of mixtures of a $4 \times 10^{-3} \text{ mol dm}^{-3}$ solution of **X** and a $3 \times 10^{-3} \text{ mol dm}^{-3}$ solution of nickel(II) chloride were measured using a colorimeter at room temperature.



What is the charge on the nickel complex?

- A** 1+ **B** 0 **C** 1- **D** 4-